

Subject Description Form

Subject Code	EEE515
Subject Title	Lasers
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject aims to provide students with a comprehensive understanding of laser principles, components, operational modes, properties, types, and applications, and equip them with the knowledge to critically analyze and make informed decisions related to laser technologies across various scientific and technological domains.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Explain laser principles: describe the fundamentals of laser operation, including stimulated emission, population inversion, and the generation of coherent light. - Analyze laser systems: differentiate between laser types, analyze operational characteristics, and understand components like gain media and resonators. - Evaluate emission modes and applications: compare laser emission modes, relate them to specific applications, and understand their significance. - Quantify gain and threshold conditions: calculate laser gain, explain threshold conditions, and understand the role of population inversion. - Design resonators and understand beam characteristics: analyze resonator designs, grasp laser beam properties, and relate them to practical applications. - Apply lasers safely and ethically: demonstrate knowledge of laser safety protocols, identify potential hazards, and consider ethical implications.

Subject Synopsis/ Indicative Syllabus	This subject covers the fundamental physics of lasers and their unique properties compared to conventional light sources. Topics include: Interaction of radiation with matter: absorption, spontaneous emission, stimulated emission, Einstein's relations. Optical resonators: stable and unstable resonators, eigenmodes, Q factor. Laser pumping: optical, electrical, and chemical pumping processes. Laser rate equations: threshold conditions, output power. Continuous wave and pulsed laser operation: Q-switching, mode-locking, ultrafast pulses. Major types of lasers: gas, solid-state, semiconductor, dye, free-electron lasers. Laser beam properties: coherence, directionality, brightness, Gaussian beams. Laser applications: materials processing, medicine, communications, sensing, scientific research.																																						
Teaching/Learning Methodology	Lecture: The fundamentals of laser physics and specific laser systems and applications will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance. Tutorial: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Assignment</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Continuous assessments consist of assignments and mid-term test. These will assess the students’ understanding of basic concepts and principles in laser technologies. The examination will be conducted to make a comprehensive assessment of students’ intended learning outcomes as stated above.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Assignment	40%	✓	✓	✓	✓	✓	✓	2. Examination	60%	✓	✓	✓	✓	✓	✓	Total	100 %						
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																			
		a	b	c	d	e	f																																
1. Assignment	40%	✓	✓	✓	✓	✓	✓																																
2. Examination	60%	✓	✓	✓	✓	✓	✓																																
Total	100 %																																						

Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs
	▪ Tutorial	6 Hrs
	Other student study efforts:	
	▪ Reading and self-study	81 Hrs
	The total student study effort	120 Hrs.
Reading List and References	Textbook • Orazio Svelto, Principles of Lasers, 5th Edition, Springer New York, NY, 2010. References • Oleksiy Shulika, and Igor Sukhoivanov, Advanced Lasers, Springer Dordrecht, 2015. • William S. C. Chang, Principles of Lasers and Optics, 1st Edition, Cambridge University Press, 2005. • Frank Träger, Springer Handbook of Lasers and Optics, 2nd Edition, Springer Berlin, Heidelberg, 2012.	